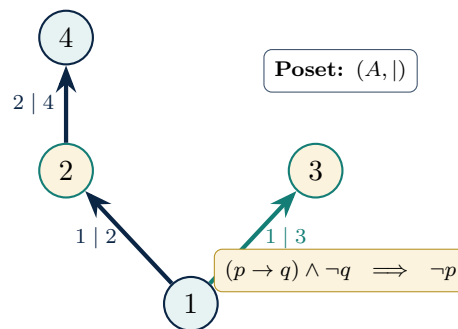


TUTORIAL TEST EXAMINATION SERIES

Term Test-01: Model Solutions & Rigorous Analysis



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ACADEMIC ID:

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COURSE:

CSE 0541 1143: Discrete Mathematics (3.00 Credits)

DEPARTMENT:

Dept. of Computer Science & Engineering

INSTITUTION:

Sylhet Engineering College (SEC)

Affiliated with Shahjalal University of Science & Technology (SUST)

★ SEC TT-01 Official Exam Specification

Exam Date: September 2026 • Full Marks: 10 • Dept. of Computer Science & Engineering

Course Code: CSE 0541 1143 • Course Title: Discrete Mathematics

Curriculum Focus: Propositional Logic (Definitions, Truth Tables) and Set Relations (Cartesian Divisibility, Reflexivity, Symmetry, Antisymmetry, Transitivity).

1 Question 01: Propositional Logic

SEC Term Test-01 — Question 01 (a)

[03 Marks]

QUESTION

Define a **proposition** and determine whether each of the following is a proposition. Give reasons.

- i) $x + 5 = 10$
- ii) Dhaka is the capital of Bangladesh.

ANSWER

1. Formal Definition of a Proposition: A **proposition** is a declarative sentence that is either strictly **True (T)** or strictly **False (F)**, but not both simultaneously.

2. Analysis of the Given Statements:

- **Statement (i):** $x + 5 = 10$
 - **Verdict:** **Not a proposition.**
 - **Mathematical Reason:** This statement contains a free variable x . Its truth value is indeterminate without context ($x = 5 \implies \mathbf{T}$, $x \neq 5 \implies \mathbf{F}$). Because its truth value is not fixed, it is an **open sentence** (or *predicate* $P(x)$).
- **Statement (ii):** “Dhaka is the capital of Bangladesh.”
 - **Verdict:** **It is a proposition.**
 - **Mathematical Reason:** It is a declarative factual assertion. Since Dhaka is indeed the capital of Bangladesh, its truth value is uniquely and definitively **True (T)**.

SEC Term Test-01 — Question 01 (b)

[02 Marks]

QUESTION

Construct the truth table for the compound proposition: $(p \rightarrow q) \wedge \neg q$

ANSWER

Since the expression contains 2 propositional variables (p, q) , there are $2^2 = 4$ truth assignments:

p	q	$p \rightarrow q$	$\neg q$	$(p \rightarrow q) \wedge \neg q$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Conclusion: The proposition $(p \rightarrow q) \wedge \neg q$ is **True** if and only if both p and q are **False**. *(This compound form serves as the antecedent of **Modus Tollens**: $[(p \rightarrow q) \wedge \neg q] \implies \neg p$)*.

2 Question 02: Relations on a Set

SEC Term Test-01 — Question 02 (a)

[02 Marks]

QUESTION

Let $A = \{1, 2, 3, 4\}$ and relation R on A defined by: $R = \{(a, b) \in A \times A \mid a \text{ divides } b\}$. Write the relation R explicitly as a set of ordered pairs.

ANSWER

a divides b (denoted $a \mid b$) implies $b = k \cdot a$ for some integer $k \geq 1$ ($b \pmod{a} = 0$). Evaluating all pairs $(a, b) \in A \times A$:

- $a = 1$: $1 \mid 1, 2, 3, 4 \implies (1, 1), (1, 2), (1, 3), (1, 4)$.
- $a = 2$: $2 \mid 2, 4 \implies (2, 2), (2, 4)$.
- $a = 3$: $3 \mid 3 \implies (3, 3)$.
- $a = 4$: $4 \mid 4 \implies (4, 4)$.

Expressing R explicitly as a set of ordered pairs:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

The relation R contains exactly **8 ordered pairs** out of $|A \times A| = 16$.

SEC Term Test-01 — Question 02 (b)

[03 Marks]

QUESTION

Determine whether R is **reflexive**, **symmetric**, **antisymmetric**, and **transitive**. Provide a brief justification for each property.

ANSWER

We evaluate each of the four fundamental relational properties on $A = \{1, 2, 3, 4\}$:

1. Reflexivity: Yes, R is Reflexive.

- *Justification:* $\forall x \in A, (x, x) \in R$ because every non-zero integer divides itself ($x \mid x$). Since $(1, 1), (2, 2), (3, 3), (4, 4) \in R$, reflexivity holds.

2. Symmetry: No, R is NOT Symmetric.

- *Justification:* $\exists x, y \in A$ such that $(x, y) \in R$ but $(y, x) \notin R$.
- *Counter-example:* $(1, 2) \in R$ (as $1 \mid 2$), but $(2, 1) \notin R$ (as $2 \nmid 1$).

3. Antisymmetry: Yes, R is Antisymmetric.

- *Justification:* $\forall x, y \in A, [(x, y) \in R \wedge (y, x) \in R] \implies x = y$. For positive integers, $a \mid b \wedge b \mid a \implies a = b$. In R , the only mutually related elements are diagonal pairs (x, x) .

4. Transitivity: Yes, R is Transitive.

- *Justification:* $\forall x, y, z \in A, [(x, y) \in R \wedge (y, z) \in R] \implies (x, z) \in R$. If $a \mid b$ and $b \mid c$, then $a \mid c$. Specifically, $(1, 2) \in R \wedge (2, 4) \in R \implies (1, 4) \in R$.

Academic Classification

Because R is **reflexive**, **antisymmetric**, and **transitive**, (A, R) constitutes a **Partially Ordered Set (Poset)**.